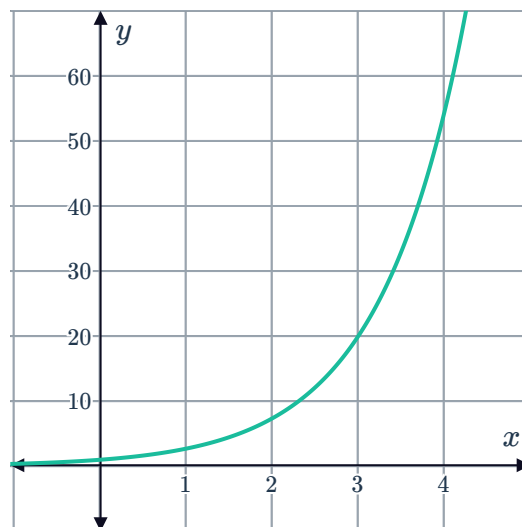


Approximate areas under graphs



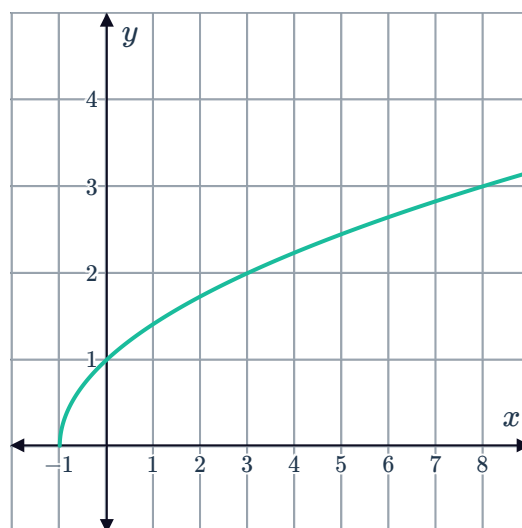
1 The following graph shows the function $y = e^x$:

- a Find an estimate for the area between the function and the x -axis for $0 \leq x \leq 4$, giving your answer to three decimal places and using:
 - i The left-endpoint approximation with 4 rectangles.
 - ii The right-endpoint approximation with 4 rectangles.
 - iii Technology and the left-endpoint approximation with 50 rectangles.
- b Use integration to find the exact area between the function and the x -axis for $0 \leq x \leq 4$.
- c Calculate the percentage error of the approximation in (a) part (iii) compared to the exact area. Round your answer to the nearest percent.



2 The following graph shows the function $y = \sqrt{x+1}$:

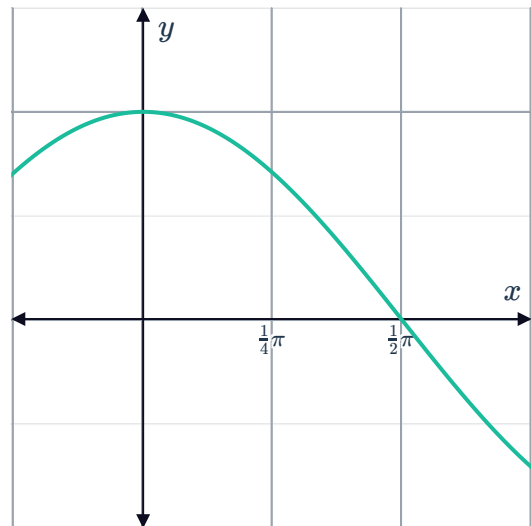
- a Find an estimate for the area between the function and the x -axis for $2 \leq x \leq 6$, giving your answer to three decimal places and using:
 - i The left-endpoint approximation with 4 rectangles.
 - ii The right-endpoint approximation with 4 rectangles.
 - iii Technology and the right-endpoint approximation with 40 rectangles.
- b Use integration to find the area between the function and the x -axis for $2 \leq x \leq 6$. Round your answer to four decimal places.
- c Calculate the percentage error of the approximation in (a) part (iii) compared to area calculated by integration in part (b). Round your answer to two decimal places.



3 The following graph shows the function $f(x) = \cos x$:

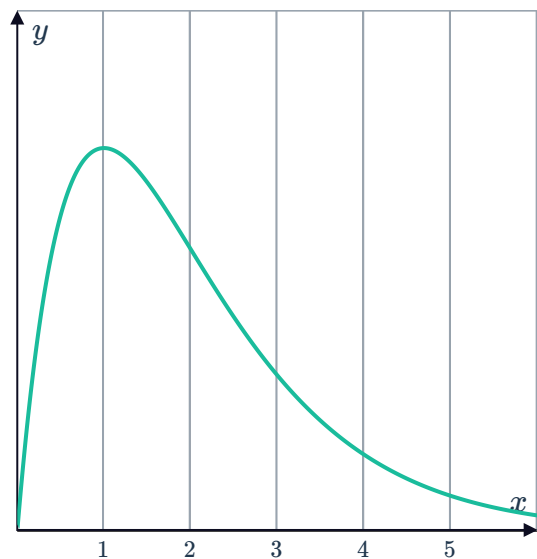


- a Find an estimate for the area between the function and the x -axis for $0 \leq x \leq \frac{\pi}{2}$, giving your answer to three decimal places and using left-endpoint approximation with:
 - i 3 rectangles.
 - ii 5 rectangles.
 - iii Technology and 50 rectangles.
- b Use integration to find the exact area between the function and the x -axis for $0 \leq x \leq \frac{\pi}{2}$.
- c Calculate the percentage error of the approximation in (a) part (iii) compared to the exact area. Round your answer to the nearest percent.



4 Consider the function $f(x) = xe^{-x}$ and the given graph of $y = f(x)$:

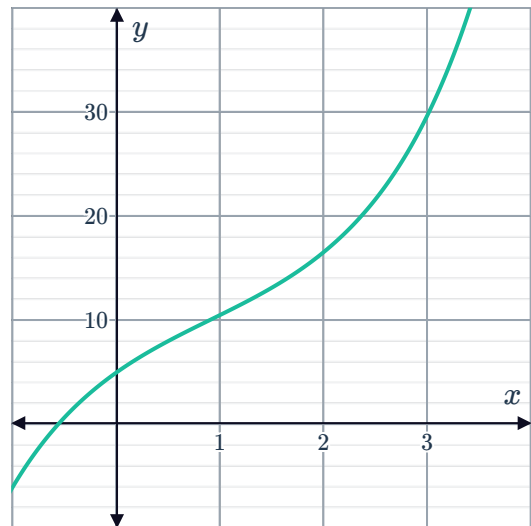
- a Find $f'(x)$.
- b Hence, show that: $\int xe^{-x} = -xe^{-x} - e^{-x} + c$.
- c Calculate an overestimate for the area bounded by $f(x)$ and the x -axis for $1 \leq x \leq 5$, giving your answer to three decimal places and using:
 - i 2 rectangles of equal width.
 - ii 4 rectangles of equal width.
- d Find the exact area bounded by $f(x)$ and the x -axis over $1 \leq x \leq 5$, using part (b).
- e Calculate the percentage error for the two estimates given in part (c). Round your answer to two decimal places.



5 Consider the function $f(x) = e^x + 10 - 6e$ and the given graph of $y = f(x)$:

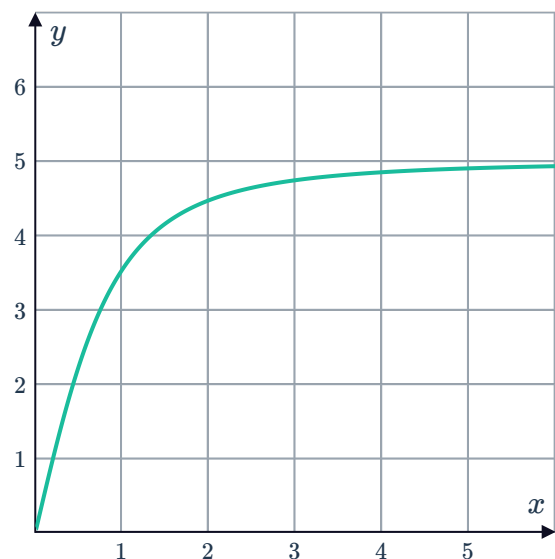


- a Calculate an overestimate for the area bounded by $f(x)$ and the x -axis for $1 \leq x \leq 3$ using 4 rectangles of equal width. Give your answer to four decimal places.
- b Calculate an underestimate for the area bounded by $f(x)$ and the x -axis for $1 \leq x \leq 3$ using 4 rectangles of equal width. Give your answer to four decimal places.
- c Given $f''(x) > 0$ for $x > 0.9$, which estimate will be closer to the exact area bounded by $f(x)$ and the x -axis for $1 \leq x \leq 3$. Explain your answer.



6 Consider the function $f(x) = \frac{5x}{\sqrt{x^2 + 1}}$ and the given graph of $y = f(x)$:

- a Calculate an overestimate for the area bounded by $f(x)$ and the x -axis for $0 \leq x \leq 2$ using 2 rectangles of equal width. Give your answer to three decimal places.
- b Calculate an overestimate for the area bounded by $f(x)$ and the x -axis for $2 \leq x \leq 4$ using 2 rectangles of equal width. Give your answer to three decimal places.
- c Which of the estimates calculated in (a) and (b) will be closer to the exact area being estimated? Explain your answer.
- d Would an underestimate or overestimate using 4 rectangles of equal width provide a closer estimate to the exact area bounded by $f(x)$ and the x -axis for $0 \leq x \leq 4$. Explain your answer.

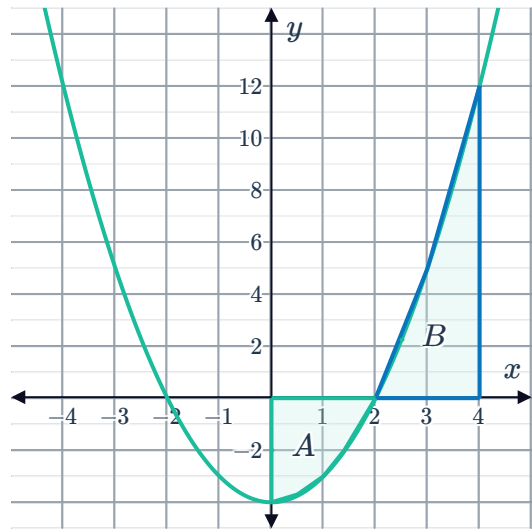




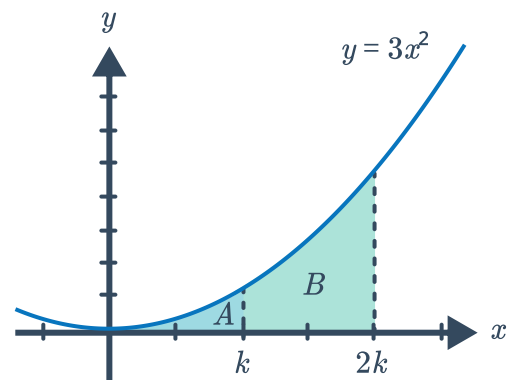
Definite integration

- 7 The following graph shows the function $y = x^2 - 4$:

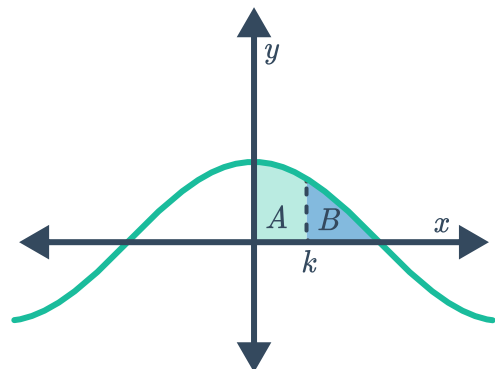
- a Find the area A .
- b Find the area B .
- c Find the ratio of the areas $A : B$.



- 8 The following graph shows the function $y = 3x^2$:
Let A define the area under the curve over the interval $[0, k]$ and B define the area under the curve over the interval $[k, 2k]$.
Find the ratio of the areas $A : B$.



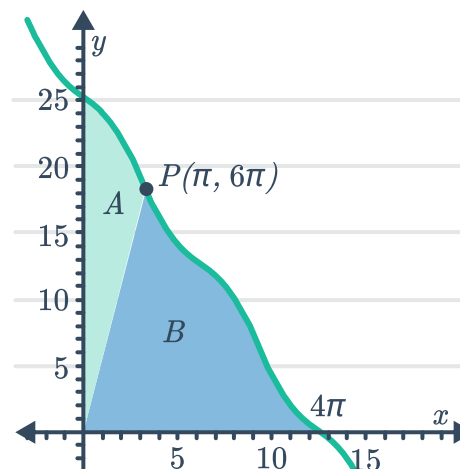
- 9 The following graph shows the function $y = \cos x$:
Find k such that ratio $A : B$ is $1 : 1$.



- 10 Consider the function $f(x) = e^x$. Let A define the area under the curve over the interval $[0, k]$ and B define the area under the curve over the interval $[0, 2k]$. Find the relationship between A and B .

- 11 The curve $y = 8\pi - 2x + \sin x$ is shown passing through the point $P(\pi, 6\pi)$. A straight line joins the origin to point P .

- Find the value of $\int_0^\pi (8\pi - 2x + \sin x) dx$.
- Find the value of A .
- Find the value of B .
- Determine the ratio $A : B$ in the form $1 : k$. Round k to two decimal places.



- 12 Consider the function $f(x) = 6ax - 6x^2$, for $a \geq 0$.

- For $a = 1$, sketch the function.
- Find the area bound by the curve and the x -axis.
- Use your calculator to complete the table of the area bound by the graph and the x -axis for different values of a .

a	1	2	3	5	10
Area (units) ²					

- Make a conjecture for the value of the area bounded by $f(x) = 6ax - 6x^2$ and the x -axis, for $a \geq 0$.
- Prove your conjecture.

13 Consider the function $f(x) = kx^n(1-x)$ on the interval $[0, 1]$, where n is a positive integer.



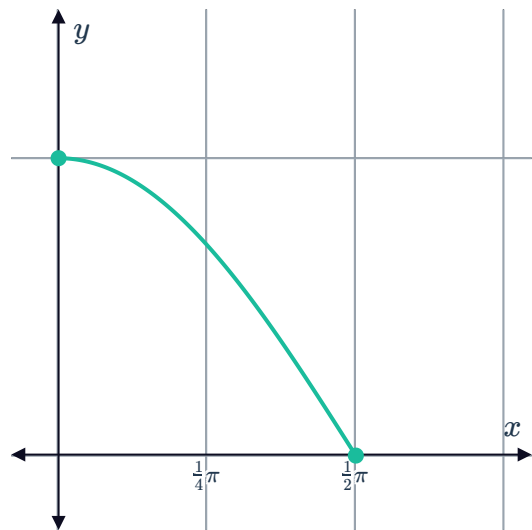
- For $n = 1$, find the value of k if the area under the curve on the interval $[0, 1]$ is 1 unit^2 .
- Using technology, find the area under the curve $y = x^2(1-x)$ on the interval $[0, 1]$.
- Hence, find the value of k that would create an area of 1 unit^2 under the curve of $f(x)$ over the interval $[0, 1]$, when $n = 2$.
- Complete the following table showing the value of k required for different values of n , if the area on the interval $[0, 1]$ is 1 unit^2 :

n	1	2	3	4	5
k					

- Make a conjecture for the value of k in terms of n , required if the area under $f(x)$ on the interval $[0, 1]$ is 1 unit^2 .
- Prove your conjecture.

14 Consider the curve $y = f_k(x)$ where $f_k(x) = \cos(kx)$ and k is a positive integer. The domain of the function $f_k(x)$ is dependent on k which is $0 \leq x \leq \frac{\pi}{2k}$.

- The given graph is of $f_1(x)$, where $k = 1$, thus $f_1(x) = \cos(1x)$ and the domain is $0 \leq x \leq \frac{\pi}{2}$. Find the area between $y = f_1(x)$ and the x -axis over the domain of the function.



- Consider $y = f_k(x)$ for $k = 2$:
 - State the function for $f_2(x)$.
 - State the domain of $f_2(x)$.
 - Find the area between $y = f_2(x)$ and the x -axis over the domain of the function.
- Find the area under $f_3(x)$ over its domain.
- Make a conjecture for the value of the area bounded by $y = f_k(x)$ and the x -axis over $0 \leq x \leq \frac{\pi}{2k}$, for any positive integer value of k .
- Prove your conjecture.